



-----IDS,IPS,Firewalls and Honeypots-----

IDS (Intrusion Detection System) :- Caught or Detect unknown traffic from internet

How IDS works :-

1. Signature Recognition :-

2. Anomaly Detection :- Malicious Behavior

3. Protocol Anomaly Detection :- Malicious Behavior on tcp/ip port

IPS (Intrusion Prevention System) :- IPS Works with IDS when IDS find Malicious Activity and detect malicious activity then IPS prevent the system from malicious activity and stop the malicious activity outside of the system.

Firewall :- stop unknown traffic or malicious traffic outside of device and system

Type of Firewalls :-

1.Hardware Firewalls :- This type of firewall basically work as a part of router and we can also implement it before router (install it before router for external)

2.Software Firewalls :- This type of firewall basically work with computer as a software based firewall

VPN (Virtual Private Network) :- use for Secure Transmission for sensitive information on untrusted internet connections (data transmission in encrypted form)

Mobile based Intrusion Detection Tools

1. zIPS
2. Wifi Inspector
3. Intruder Detector

Mobile based Firewall :-

1. Mobiwol Noroot Firewall
2. Privacy Shield
3. NetPatch Firewall

HoneyPot :- honeypot is a tool used for attract attacker and caught who is attack on you. HoneyPot is generate fake port on our system and when an attacker attack on our system they can easily caught by us.

Honeybot Download

What is a Web application firewall(WAF) -

It is a technique to provide an extra security layer over Web Applications via the configuration of a Web Application Firewall [WAF].

Like - such as Cloudflare, ASPA Firewall, FirePass, FortiWeb, Huawei Cloud Firewall, HyperGuard, WTS-WAF, etc.

Wafw00f is a popular Python program/tool that takes the guesswork of fingerprinting a website's firewall off your hands. Wafw00f can determine the underlying firewall used by a service that it probes based on the responses to a

series of carefully crafted web requests It is used to identify the WAF over web applications.

Download Link:- <https://github.com/EnableSecurity/wafw00f>

Installation :

```
#git clone https://github.com/EnableSecurity/wafw00f.git
```

```
#cd wafw00f
```

```
#chmod 777 *
```

```
#python setup.py install
```

Commands :

```
#wafw00f -l
```

```
#wafw00f https://www.domain.com
```

```
#wafw00f -a https://www.domain.com
```

```
#wafw00f -v https://www.domain.com
```

Nmap: This tool is used to perform many things over here.

Commands :

```
#nmap -p 80,443 <ip address/ domain.com>
```

```
#nmap -p 80,443 --script=http-waf-detect <ip address/ domain.com>
```

```
#nmap -p 80,443 --script=http-waf-fingerprint <ip address/ domain.com>
```